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APPLICATION OF SOFT COMPUTING

Coverage path planning for multiple UAVs in prioritized regions using ant colony optimization

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Abstract
The increasing number of maritime transport vessels has elevated ship distress accidents, emphasizing the imperative for effective Search and Rescue (SAR) operations. In this study, we introduce a coverage path planning framework for multiple Unmanned Aerial Vehicles (UAVs) to optimize maritime SAR missions by explicitly accounting for regional priorities. The proposed framework hierarchically incorporates these priorities and derives coverage paths using a heuristic algorithm based on Ant Colony Optimization (ACO) for each priority region. The algorithm consists of two key phases: the Region Allocation Phase (RAP), which utilizes a greedy approach guided by regional importance, and the Dynamic Refinement Phase (DRP), which employs ACO to ensure balanced resource usage. Comprehensive numerical experiments, utilizing actual sea coordinates, are conducted to validate the effectiveness of the proposed methodology. The proposed method outperforms both a MILP model and a reinforcement learning-based approach, particularly in large-scale instances. It demonstrates computational robustness and scalability, consistently generating high-quality solutions within practical computation times, even in complex environments. The methodology is further validated through experiments on irregular environments, confirming its adaptability and effectiveness.

Keywords Maritime search and rescue · Unmanned Aerial Vehicle (UAV) · Priority-based coverage path planning · Ant Colony Optimization (ACO)

1 Introduction

With the development of the global ocean economy, ocean-based industries such as marine leisure and maritime transportation have been growing rapidly (Woo et al. 2024; Lee and Cho 2025). As the quantity of maritime transport vessels rises, there is a gradual increase in the occurrence of ship distress accidents (Li et al. 2020). Since 2018, the IMO Global Integrated Shipping Information System (GISIS) has recorded a total of 1,141 incidents over the past five years, comprising 752 cases classified as very serious casualties and 261 cases categorized as serious casualties (IMO 2023). To address these challenges, it is important for the maritime authority to provide effective maritime Search and Rescue (SAR) services. The primary objective of maritime SAR operations is the prompt and precise search for and rescue of the distressed individuals, which significantly influences the survival rate of the victims (Matos et al. 2013). To enhance the efficiency of maritime SAR operations, it is crucial to ensure effective coordination and decision-making by maritime authorities and maximize the utilization of available resources. Furthermore, the dynamic nature of the marine environment can significantly impact the outcome of these operations (Coppini et al. 2016; Xiong et al. 2020; Nanda et al. 2024).

Maritime SAR operations consist of two distinct levels: operational decision-making and strategic planning

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(Siljander et al. 2015). At the operational level, the objective is to detect missing objects with minimal time for real-time SAR operations, considering stochastic environmental characteristics such as waves, wind, tides, and currents (Frost and Stone 2001; Breivik and Allen 2008; Razi and Karatas 2016; Brushett et al. 2017; Motamedi et al. 2024). At the planning level, the objective is to assess the overall effectiveness of the SAR response system. This involves achieving maximum coverage of a sea area while utilizing the minimum necessary number of rescue units. Deployment of resources (helicopters, lifeboats, fixed-wing aircraft, unmanned aerial vehicles, etc.) and the initial search operations are strategically planned in advance to ensure efficient SAR operations. The resource allocation problem aims to use historical accident data to identify locations in close proximity to where accidents are likely to occur and to determine the number of resources to be deployed at each location (Abi-Zeid and Frost 2005; Azofra et al. 2007; Wagner and Radovilsky 2012; Ai et al. 2015; Pelot et al. 2015; Siljander et al. 2015; Karatas et al. 2017; Akbari et al. 2018; Xiong et al. 2020; Choi et al. 2025). In the initial search of maritime SAR operations, Coverage Path Planning (CPP) is crucial for systematically exploring a workspace by generating valid paths to cover all areas with the shortest search path (Ai et al. 2021; Cai et al. 2022). In this paper, we focus on CPP for initial search operations at the planning level in maritime SAR operations.

In the maritime domain, the development of SAR resources is crucial due to the limitation of direct human access to the sea following accidents. Relying solely on traditional resources often results in operational shortcomings, resulting in reduced SAR efficiency and prolonged response times. To address these challenges, The International Maritime Rescue Federation is enhancing their maritime SAR capabilities, incorporating robotic technologies (Cubber et al. 2017). These robots offer significant advantages, facilitating swift responses to marine disasters and demonstrating the ability to operate continuously without rest in environments (Otto et al. 2018). With the rapid progress in automation and artificial intelligence, Unmanned Aerial Vehicles (UAVs) have found widespread applications in diverse fields such as traffic monitoring, disaster relief, and surveillance. UAVs outperform manned aircraft in navigating hazardous conditions and conducting repetitive tasks, enhancing the efficiency of SAR operations. Specifically designed for maritime SAR, UAVs possess advantages including agility, portability, and air accessibility, enabling them to cover vast areas efficiently with fewer obstacles (Lin and Goodrich 2009; Lin et al. 2010; Keller et al. 2016; Chen et al. 2022). UAVs equipped with remote sensing devices are extensively employed for SAR purposes during disasters like earthquakes, landslides, and shipwrecks (Adams and Friedland 2011; Jorge et al. 2019).

In particular, due to the constraints associated with a single UAV, such as limited search range and battery capacity during the SAR operations, studies on multiple UAVs operations are increasingly emphasized. The utilization of multiple UAVs offers significant advantages in achieving rapid full coverage. However, it introduces additional considerations, including cooperation types, information sharing, task allocation, and collision avoidance (Almadhoun et al. 2019). Several studies have explored the implementation of multiple UAVs in maritime SAR operations (Lee and Morrison 2015; Dinnbier et al. 2017; Alotaibi et al. 2019; Queralta et al. 2020). Furthermore, research has focused on the application of multiple UAVs in the CPP for maritime SAR, aiming to minimize completion time (Cho et al. 2021, 2022). However, due to the dynamic nature of the sea and the movement of victims with ocean currents, swift access to the accident site is crucial to minimize uncertainty in locating survivors over time. Given the urgency of search operations and the characteristics of maritime environments, it is imperative to consider the priority of the search area for efficient coverage of target regions.

This study aims to propose an efficient search path method for multiple UAVs in maritime SAR operations, motivated by the critical need to minimize the time required to locate survivors in life-threatening situations. The focus is on minimizing search time while emphasizing the prioritization of search areas. The proposed CPP framework for multiple UAVs takes into account the regional priorities of the search area, addressing the limitation of many existing CPP approaches that often treat all regions equally without considering their relative importance. The proposed framework allows survivors to be located and tracked quickly by searching the areas with a higher likelihood of survivor presence. A heuristic algorithm based on Ant Colony Optimization (ACO) is proposed. This algorithm generates search paths considering heterogeneous UAVs by learning pheromones reflecting the characteristics of UAVs. In summary, the contributions of this study are threefold: (i) explicitly incorporating regional priorities into path planning to enhance search effectiveness, (ii) accounting for heterogeneous UAV capabilities such as speed and endurance for more realistic task allocation, and (iii) developing an ACO-based heuristic that evaluates and refines solutions to provide near-optimal search paths within practical computation time. The following sections present the formal problem description, our proposed methodology, and experimental validation.

The remainder of this paper is organized as follows. Section 2 provides a comprehensive overview of prior research. Section 3 presents the problem description of the study.

Section 4 introduces the proposed methodology. Subsequently, Section 5 describes numerical experiments conducted using empirical data. Lastly, Section 6 draws the conclusions of the research.

2 Related works

CPP holds significant importance in enhancing the efficiency of UAVs in executing search operations by systematically traversing all regions of interest (Choset 2001; Galceran and Carreras 2013; Chen et al. 2022; Kumar and Kumar 2023).

The coverage paths are influenced by the shape of the area, ranging from regular to irregular geometries. Various decomposition methods, including exact and approximate cellular decomposition, are used to handle different complexities and shapes in the region, as discussed by several authors (Choset 2001; Galceran and Carreras 2013; Cabreira et al. 2019; Fevgas et al. 2022; Kumar and Kumar 2023). Cellular decomposition methods, including exact methods like trapezoidal decomposition (Latombe and Latombe 1991), boustrophedon decomposition (Choset 2000), and morse-based decomposition (Acar et al. 2002), are crucial for simplifying region geometry based on both region characteristics and UAV capabilities such as speed and battery duration (Kumar and Kumar 2023). In contrast, approximate cellular decomposition or grid-based decomposition overlooks region characteristics and divides the area into grids with a fixed unit size, introducing variable grid shapes and potential variations in UAV movement paths. While cellular decomposition considers region and UAV characteristics, it can struggle in complex scenarios, often yielding suboptimal solutions. On the other hand, grid-based decomposition generates optimal paths but at the cost of higher computational complexity (Valente et al. 2013; Wu et al. 2024). For SAR missions requiring rapid response and optimal multiple UAVs solutions in situations with limited region information, approximate cell decomposition is suitable. This study specifically employs approximate cellular decomposition (grid-based decomposition), focusing on investigating decomposition angles rather than shapes, considering advancements in UAV movement capabilities (Feng et al. 2021).

Research on CPP has gained attention in both single UAV (Artemenko et al. 2016; Li et al. 2016; Bouzid et al. 2017; Coombes et al. 2017; Cabreira et al. 2018) and multiple UAVs (Nam et al. 2016; Paradzik and Ince 2016; Albani et al. 2017; Balampanis et al. 2017). The use of a single UAV for extensive coverage presents challenges related to time, distance, energy, and information quality (Galceran and Carreras 2013). While multiple UAVs CPP strategies align with single UAV coverage principles, additional considerations such as cooperation type, information sharing, autonomy level, and task allocation must be addressed. Moreover, multiple UAVs are categorized into homogeneous and heterogeneous types. Homogeneous UAVs share identical features, promoting streamlined coordination but potentially limiting adaptability. In contrast, heterogeneous UAVs constitute a diverse fleet with varying characteristics, offering greater flexibility and visibility for different tasks but introducing complexities in communication and coordination. Nevertheless, it is crucial to highlight that research on heterogeneous UAVs remains limited (Almadhoun et al. 2019). This study aims to address this gap by specifically focusing on CPP for heterogeneous multiple UAVs.

Previous studies have utilized Mixed-Integer Linear Programming (MILP) to tackle the CPP problem involving multiple UAVs, striving to achieve efficient and optimal solutions in various applications. In the context of maritime SAR missions, Cho et al. (2021, 2022) crafted a MILP model tailored for optimal CPP within grid-decomposed regions. The aim was to minimize the maximum completion time, ensuring efficient coordination among multiple UAVs. Yet, in SAR scenarios, the main objective is prompt target rescue. When information about potential target locations, like probabilities or weights, is available, it is advisable to prioritize path planning for areas with the highest likelihood of harboring the target. Hong et al. (2021) formulated a MILP model that maximized the total time-weighted probability of target discovery within known subareas. However, when facing time-critical decisions, focusing solely on one objective may not yield satisfactory results (Gosiewski and Kwaśniewski 2020). Therefore, Saha et al. (2021) introduced a MILP model within a multi-objective framework. Their first objective aimed to maximize the cumulative probability of target detection, while the second aimed to minimize cumulative completion time. The study transformed multi-objectives into a single objective with weighted sums to achieve real-time solutions in time-sensitive missions like SAR. Additionally, Kazemdehbashi and Liu (2025) proposed a fast algorithm for planning multi-UAV flight paths in windy conditions to efficiently search rectangular areas. By formulating the problem as a MIP model, the method ensures near-optimal coverage with low computational cost.

Despite extensive studies on multi-UAV CPP, two critical limitations remain in the literature. First, most approaches either treat all regions with equal importance or use simple probabilistic weighting, which is insufficient for missions where time-sensitive regional prioritization is essential. Second, while multiple UAVs are often considered, the majority of existing models assume homogeneous UAVs and thus fail to capture differences in speed, endurance, or

energy capacity. These gaps reduce the effectiveness of CPP in realistic, mission-critical settings such as maritime SAR.

The main contributions of this paper can be outlined as follows: First, unlike existing approaches such as Hong et al. (2021); Saha et al. (2021); Kazemdehbashi and Liu (2025), which focus primarily on minimizing search time or maximizing coverage probabilities, our method explicitly incorporates the priority of search areas into path planning. Whereas most studies treat all regions equally or rely only on probabilistic weights, our model assigns higher priority to critical regions, thereby enhancing search effectiveness in time-sensitive missions. Second, although multi-UAV CPP has been widely studied (Valente et al. 2013; Acevedo et al. 2013; Saha et al. 2021), most methods—including Kazemdehbashi and Liu (2025)—assume homogeneous UAVs and overlook differences in speed, endurance, and other capabilities. Our model accounts for heterogeneous UAV characteristics, leading to more efficient and realistic task allocation. Third, while the hierarchical two-phase framework of grid map construction and path planning has been commonly adopted in CPP literature, our contribution lies in embedding the priority-aware path planning into this framework through an ACO-based algorithm. This integration enables near-optimal solutions within practical computation time, providing both scalability and methodological distinctiveness.

3 Problem description

The objective of this study is to enhance the efficiency of search operations in maritime SAR systems. In the initial search operations, CPP plays a critical role in thoroughly searching the area surrounding the accident site and locating victims as quickly as possible. As illustrated in Fig. 1, the search area is characterized by varying regional priorities based on proximity to the accident point. To improve search effectiveness, paths should be generated first for regions with a higher likelihood of survivor presence. Accordingly, this study proposes a CPP framework that generates search paths sequentially based on regional priority levels. This framework consists of two primary stages. First, the search area is divided into a discrete grid, which transforms the continuous geographic space into a structured graph problem. Second, coverage paths are generated over this graph.

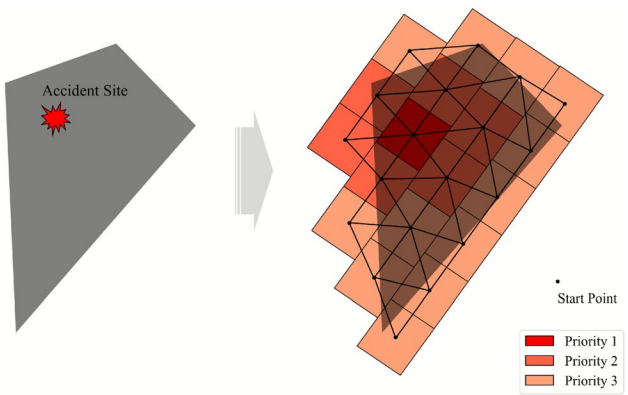


Fig. 1 Graph-based search area with regional priorities

In the first stage, we adopt the grid-based area decomposition method proposed by Cho et al. (2021). This method decomposes the target area into square-shaped grid cells based on the camera footprint of the UAV, transforming the search area into a graph with vertices and edges while considering the angle of decomposition. This approach is proven to minimize the size of the decomposed coverage area. Specifically, the grid-based area decomposition procedure aims to find the optimal orientation for decomposition that minimizes the total area covered by the grid cells. This process is performed iteratively for each edge k of the search area polygon. For a given edge k:

(1) The area is conceptually sliced into parallel layers $l \in L_k$, where each layer has a height h_k corresponding to the camera footprint dimension a.
(2) Within each layer l, the coordinates of the top-left (c_l^{LT}), bottom-left (c_l^{LB}), top-right (c_l^{RT}), and bottom-right (c_l^{RB}) boundary points relative to the layer's orientation are determined.
(3) The actual leftmost point $p_k^L(l)$ and rightmost point $p_k^R(l)$ are calculated for each layer l. This calculation uses the layer's boundary coordinates and accounts for any polygon vertices v_k that might intersect the layer. The formal calculations are provided in Equations (1) and (2). Here, $I_l^{V_k}$ is an indicator variable (1 if vertex v_k is in layer l, 0 otherwise), c^{v_k} is the coordinate of vertex v_k, and M is a large number used to handle boundary conditions defined by vertices (Cho et al. 2021).

$$p_k^L(l) = \min_x \{c_l^{LB}, c_l^{LT}, \{(1 - I_l^{V_k})M + c^{v_k} : v_k \in V\}\} \quad (1)$$

$$p_k^R(l) = \max_x \{c_l^{RB}, c_l^{RT}, \{I_l^{V_k} c^{v_k} : v_k \in V\}\} \quad (2)$$

(4) The number of grid cells (nodes) $R(l)$ required to cover the width of layer l is determined by dividing the distance between the leftmost and rightmost points by the camera footprint size a, taking the floor of the result: $R(l) = \lfloor |p_k^R(l) - p_k^L(l)|/a \rfloor$.
(5) The coordinates $p_k^n(l)$ for each of the n nodes within a layer are calculated along the line segment between $p_k^L(l)$ and $p_k^R(l)$. These nodes are placed equidistantly, oriented by the unit vector u^k of edge k, and are

bounded by the following normalization condition in Equation (3).

$$p_k^n(l) = p_k^L(l) + \left((n-\tfrac{1}{2})u^k\right)\left(\frac{|p_k^R(l) - p_k^L(l)|}{R(l)}\right), \quad 0 \le \tfrac{n-\frac{1}{2}}{R(l)} \le 1. \tag{3}$$

(6) The total effective decomposed area z_k generated when decomposing relative to edge k is calculated by summing the widths $|p_k^R(l) - p_k^L(l)|$ across all layers $l \in L_k$ and multiplying by the layer height h_k, as shown in Equation (4).

$$z_k = h_k \times \sum_{l \in L_k} |p_k^R(l) - p_k^L(l)| \tag{4}$$

This procedure is repeated for all edges k of the polygon. The decomposition orientation (defined by the edge k) that yields the minimum total area z_k according to Equation (4) is selected. The set of all node coordinates $p_k^n(l)$ generated by this optimal decomposition constitutes the final set of nodes N^{IN} used for the path planning phase. The arcs (edges) of the resulting graph are implicitly defined by the adjacency relationship between these generated nodes (grid cells).

Once the search area is represented as a grid graph, regional priorities are assigned to guide the search. These priorities, which model a survivor probability map, are defined using a concentric layer model based on grid adjacency. The priority level $P(c)$ for any given cell c is formally determined by its graph distance from the cell containing the incident point, $c_{incident}$:

$$P(c) = \begin{cases} 3 \text{ (High)} & \text{if} \quad 0 \le d(c, c_{incident}) \le k_H \\ 2 \text{ (Medium)} & \text{if} \quad k_H < d(c, c_{incident}) \le k_M \\ 1 \text{ (Low)} & \text{if} \quad d(c, c_{incident}) > k_M \end{cases} \tag{5}$$

Here, $d(c, c_{incident})$ represents the graph distance, calculated as the minimum number of cell-to-adjacent-cell steps required to travel from cell c to the incident cell. The integer thresholds k_H and k_M define the thickness of each concentric priority band.

Based on the graph representation derived from the area decomposition and the stated assumptions, the multi-UAV CPP problem is formulated as a MILP model using the notations in Table 1. The goal is to find paths for each UAV that collectively cover all designated nodes (grid cells) within the search area while minimizing the overall mission completion time, defined as the time taken by the UAV that finishes last.

To develop a CPP framework for multiple UAVs, taking into account regional priorities with a heightened probability of survivor presence in maritime SAR missions, the following assumptions are introduced:

(1) Each UAV is located at a specific node (e.g., starting point) at the start of the route.
(2) UAVs prioritize searching for higher-priority areas. In this study, the definition of search priorities (e.g., based on survivability probability) is assumed to be given externally.
(3) UAVs move only to adjacent nodes from their current location. However, when moving from a higher-priority search area to a lower-priority area, they can move to any node.
(4) UAVs move at a constant speed and require additional time to change the angle of the route.
(5) UAVs have limited flight time due to battery constraints.

Based on the graph representation derived from the area decomposition and the stated assumptions, the multi-UAV CPP problem is formulated as a MILP model using the notations in Table 1. The goal is to find paths for each UAV that collectively cover all designated nodes (grid cells) within the search area while minimizing the overall mission completion time, defined as the time taken by the UAV that finishes last.

The mathematical model is formulated as:

$$\min \quad z = \max_{m \in U} t_m \tag{6}$$

Table 1 Notations

Sets	
N	Set of nodes, including a starting node i^{start} outside the search area
N^{IN}	Set of nodes inside the actual search area ($N = N^{IN} \cup \{i^{start}\}$)
U	Set of UAVs, indexed by m
S	Set of sequence steps for nodes in a path, indexed by s
$N^{ADJ}(i)$	Set of nodes adjacent to node i
Parameters	
d_{ij}	Distance between the centers of node i and node j
φ_{ijk}	Angle of rotation for UAV moving along path segment $i \to j \to k$
v_m	Maximum cruising speed of UAV m
ω_m	Rotational speed (angular velocity) of UAV m
e_m	Endurance (maximum flight time) of UAV m
Decision Variables	
x_{ijkms}	1 if UAV m covers path sequence $i \to j \to k$ at step s; 0 otherwise
t_m	Total completion time for UAV m

s.t.

$$\sum_{i\in N}\sum_{j\in N}\sum_{k\in N} x_{ijkms} = 1, \qquad \forall m \in U, s \in S \tag{7}$$

$$\sum_{j\in N^{ADJ}(i^{start})}\sum_{k\in N^{ADJ}(j)} x_{i^{start}jkms} = 1, \qquad \forall m \in U, s = 1 \tag{8}$$

$$\sum_{i\in N}(x_{ijkms} - x_{jikm(s-1)}) = 0, \qquad \forall j,k \in N, m \in U, s \in S \setminus \{1\} \tag{9}$$

$$\sum_{j\in N}\sum_{k\in N}\sum_{m\in U}\sum_{s\in S} x_{ijkms} = 1, \qquad \forall i \in N^{IN} \tag{10}$$

$$t_m = \sum_{i\in N}\sum_{j\in N}\sum_{k\in N}\sum_{s\in S}\left(\frac{d_{jk}}{v_m} + \frac{\varphi_{ijk}}{\omega_m}\right) x_{ijkms}, \qquad \forall m \in U \tag{11}$$

$$t_m \le e_m, \qquad \forall m \in U \tag{12}$$

$$t_m \ge 0, \qquad \forall m \in U \tag{13}$$

$$x_{ijkms} \in \{0,1\}, \qquad \forall i,j,k \in N, m \in U, s \in S \tag{14}$$

The objective function (6) minimizes the maximum UAV completion time t_m. Constraint (7) ensures that each UAV selects exactly one sequence $i \to j \to k$ at each step. Constraint (8) specifies that each UAV's path begins from the starting node i^{start} at step $s = 1$. Constraint (9) ensures path connectivity across sequence steps. Constraint (10) guarantees that each internal node is visited exactly once as the starting point of a sequence. Constraint (11) calculates the total time t_m as the sum of travel and turning times, and Constraint (12) ensures that t_m does not exceed the UAV's endurance. Finally, Constraints (13) and (14) define the domains for completion times and binary path selection variables.

Solving this MILP model guarantees an optimal set of paths. However, the model belongs to the class of NP-hard problems, meaning its required computation time grows exponentially with the number of nodes and UAVs. This complexity makes it intractable for the large-scale, time-sensitive scenarios found in real-world SAR missions. To overcome this critical limitation, an efficient heuristic is required to generate near-optimal solutions within a practical timeframe. Therefore, the subsequent section introduces our novel heuristic approach, designed to balance solution quality with computational efficiency for realistic problem instances.

4 Proposed methodology

The proposed Dynamic Allocation Algorithm is structured into two distinct phases: the Region Allocation Phase (RAP) (Phase 1) and the Dynamic Refinement Phase (DRP) (Phase 2). These two phases are executed sequentially for each weighted area, beginning with regions of higher priority (i.e., higher weight) and proceeding to lower-priority ones. Detailed descriptions of each phase are provided in the subsequent sections.

In the RAP, the algorithm generates an initial feasible allocation of UAVs to designated regions. During this allocation process, a Path Repair function is employed to resolve potential deadlocks or infeasible paths, thereby ensuring that the generated allocations are valid and can serve as a strong foundation for later optimization. This phase establishes a baseline solution from which further improvements can be pursued.

Following the RAP, the DRP begins. In this phase, the algorithm uses ACO to dynamically adjust both region assignments and the UAV paths within them. The primary goal of the DRP is to correct unbalanced workloads. It achieves this by reassigning areas and optimizing paths, which improves both overall efficiency and load balancing.

A full iteration of the algorithm consists of applying both RAP and DRP across all weighted regions, thereby generating a complete set of routes for the entire surveillance area. This process is repeated ten times, and the best overall solution is selected based on the minimum objective value (i.e., makespan). During each iteration, the algorithm maintains a historical record of the best objective values. This record is kept for each weighted area across both the RAP and DRP phases. These records serve as baselines for evaluating whether the current solution offers meaningful improvement.

To enhance computational efficiency, an early termination mechanism is implemented. Specifically, during the RAP, if the current objective value for a weighted area is worse than the best value recorded in prior iterations, the algorithm immediately halts the current iteration and proceeds to the next one. A similar rule applies in the DRP: if the refined objective value does not surpass the previous minimum for the area, the iteration is terminated early. These early stopping criteria help prevent unnecessary computation and allow the algorithm to focus on more promising solution paths.

The overall structure and logic of the Dynamic Allocation Algorithm are summarized in Algorithm 1.

Input: Region set $\mathcal{R}$, Priority set $\mathcal{W}$, UAV set $\mathcal{U}$
Output: Optimized Path set $\tilde{P}_{\mathcal{U}}$

Initialize Stopping Criteria $\mathcal{C}_{\mathcal{I}}^{\mathcal{W}}$
for iteration = 1 to 10 **do**
  for w in $\mathcal{W}$ **do**
    Initialize Path Set $P_{\mathcal{U}}^{w}$
    while $RAP(\mathcal{R}_i^w, \mathcal{U})$ **do**
      $T_{\mathcal{U}}^{w} \leftarrow$ tour length of $P_{\mathcal{U}}^{w}$
      if $T_{\mathcal{U}}^{w} > \mathcal{C}_1^w$ **then**
        Skip to Next Iteration
      end if
    end while
    $P_{\mathcal{U}}^{w} \leftarrow RAP(\mathcal{R}_i^w, \mathcal{U})$
    $\tilde{P}_{\mathcal{U}}^{w} \leftarrow DRP(P_{\mathcal{U}}^{w})$
    $\tilde{T}_{\mathcal{U}}^{w} \leftarrow$ tour length of $\tilde{P}_{u}^{w}$
    if $\tilde{T}_{\mathcal{U}}^{w} > \mathcal{C}_2^w$ **then**
      Skip to Next Iteration
    end if
    Update $\mathcal{C}_1^w$ and $\mathcal{C}_2^w$
  end for
end for

Algorithm 1 Dynamic Allocation Algorithm

4.1 Region Allocation Phase (RAP)

In this phase, initial regions are allocated to UAVs through the iterative implementation of the RAP for each priority level. RAP generates a solution by employing a greedy approach, selecting the neighbor node with the minimum travel time from the current node.

4.1.1 Roulette wheel selection

In the RAP, we employ a roulette wheel selection process to assign regions to available UAVs. The assignment criteria prioritize UAVs with the minimum travel time at each allocation step of the RAP procedure. This process involves two primary steps:

(1) **UAV Selection**: A UAV, denoted as $u_{\min}$, is selected based on the criterion of minimum travel time. The current position of this UAV is at region R_i, which is the terminal node of its current path.
(2) **Next Node Selection**: The subsequent node for $u_{\min}$ is determined through Roulette Wheel Selection, based on the candidate nodes adjacent to current region R_i . These candidate nodes, $\mathcal{C}(u_{\min})$, consist of all adjacent, unvisited regions reachable within $u_{\min}$'s battery duration.

The battery duration time of UAV u, $\mathcal{B}_u$ is calculated as follows:

$$\mathcal{B}_u = \mathcal{F}_u - \frac{d_{\text{tot}}}{v_{\max,u}} - \frac{\theta_{\text{tot}}}{r_u} \tag{15}$$

Here, $\mathcal{F}_u$ denotes its maximum flight time, $v_{\max,u}$ is its maximum velocity, and r_u indicates its rotation speed. The variables d_{tot} and θ_{tot} refer to the total distance and total rotation angles, respectively, which are the cumulative sums of distances and rotation angles along the current path.

For the selection of the next region, we calculate the flight time $TF_{j,k}$ of u_{min} to each candidate region R_k from R_j:

$$TF_{j,k} = \frac{d_{j,k}}{v_{\max,u_{min}}} + \frac{\theta_{i,j,k}}{r_{u_{min}}} \tag{16}$$

Here, *i* represents the penultimate region in the current path. The variables $d_{j,k}$ and $\theta_{i,j,k}$ refer to the distance from region R_j to R_k and rotation angles of the region R_i, R_j and R_k, respectively. The Roulette Wheel algorithm favors regions with lower path times, enhancing efficiency.

4.1.2 Path repair process

During region allocation, a deadlock may occur where all UAVs lack candidate nodes for allocation, even with unassigned regions remaining. To resolve this, we initiate a Path Repair function. This function determines a base node for re-pathing based on the path developed thus far. The repair process involves:

(1) Maintaining the original path up to the selected base node.
(2) Re-pathing the subsequent path in reverse order to form the final repair path.

Given a deadlock path P, we define the base node for the repaired path of Unmanned Aerial Vehicle (UAV) i, denoted as $B_{i,k}$, through a softmax function applied to the lengths of unvisited adjacent nodes of the terminal node of the initial path. Specifically, the selection probability for a candidate base node k is given by:

$$B_{i,k} = \frac{e^{|A(k)\cap U|}}{\sum_{j\in A(P_{-1})\setminus\{P_{-2}\}} e^{|A(j)\cap U|}} \tag{17}$$

Here, $A(k)$ represents the set of nodes adjacent to node k, and U denotes the set of unvisited nodes. Furthermore, P_{-n} symbolizes the nth last node of the path P. This approach facilitates the UAV in selecting a new base node that optimally navigates away from the deadlock while prioritizing unvisited regions. This approach prioritizes nodes with more unvisited adjacent regions, thus enhancing the exploration efficiency. The number of repair attempts is limited to

the hyperparameter p, and if these attempts fail to resolve the dead-lock, the region allocation process is initiated.

The detailed algorithms for each phase of the Region Allocation process are outlined in Algorithms 2, 2-1 and 2-2, respectively. This structured approach ensures a comprehensive and efficient region allocation for UAVs in varied operational environments.

Input: Region set $\mathcal{R}$, UAV set $\mathcal{U}$
Output: Region-UAV allocation $\mathcal{A}$

$\mathcal{R}_{\text{ua}} \leftarrow \mathcal{R}$ /* *Initialize unassigned regions* */
$\mathcal{A}, \mathcal{T} \leftarrow (\emptyset, 0)$ for each $u \in \mathcal{U}$ /* *Assignments and finish times* */
$\mathcal{C} \leftarrow \emptyset$ /* *Initialize candidate nodes* */
while $\mathcal{R}_{\text{ua}} \neq \emptyset$ **do**
  for u in $\mathcal{U}$ **do**
    $i, j \leftarrow$ Retrieve last two nodes from u's path
    $\mathcal{C}(u) \leftarrow$ Nodes adjacent to j within u's remaining battery
  end for
  if $\mathcal{C}$ is empty **then**
    Attempt path repair up to p times; if fail, restart allocation
  else
    $u_{\min} \leftarrow$ Select UAV with minimum travel time to a node in $\mathcal{C}$
    $k \leftarrow$ Choose next region for $u_{\min}$ via roulette wheel from $\mathcal{C}(u_{\min})$
    $\mathcal{R}_{\text{ua}} \leftarrow \mathcal{R}_{\text{ua}} \setminus \{k\}$
    $\mathcal{A}(u_{\min}) \leftarrow \mathcal{A}(u_{\min}) \cup \{k\}$
    $\mathcal{T}(u_{\min}) \leftarrow \mathcal{T}(u_{\min})+$ Travel time to R_k
    Update $u_{\min}$'s battery $\mathcal{B}_{u_{\min}}$ according to travel
  end if
end while

Algorithm 2 Region Allocation Phase (RAP)

Input: Selected UAV $u_{\min}$, Candidate nodes $\mathcal{C}(u_{\min})$
Output: Selected node c^*

$j \leftarrow$ last node visited by $u_{\min}$
$S \leftarrow$ Sequence $\{e^{1/TF_{j,c}} : c \in \mathcal{C}(u_{\min})\}$
$P \leftarrow$ cumulative sums of S
$r \leftarrow$ random number in $[0, P_{\text{end}}]$ /* P_{end} *is the last element of* P */
for $i = 1$ to $|\mathcal{C}(u_{\min})|$ **do**
  if $r \leq P(i)$ **then return** $\mathcal{C}(u_{\min})(i)$ /* *Return the* i*-th node in* $\mathcal{C}(u_{\min})$ */
  end if
end for

Algorithm 2-1 Roulette Wheel Selection for UAV Path

Input: A deadlock path P, Unvisited node set U
Output: Repaired path R

for each node k in $A(P_{-1}) \setminus \{P_{-2}\}$ **do**
  compute $B_{i,k} = \frac{e^{|A(k) \cap U|}}{\sum_{j \in A(P_{-1}) \setminus \{P_{-2}\}} e^{|A(j) \cap U|}}$
end for
$c \leftarrow \text{argmax}_k B_{i,k}$
$R \leftarrow$ *concat*(initial node of P up to c, and P_{-1} to c)
return R

Algorithm 2-2 UAV Path Repair Function

4.2 Dynamic Refinement Phase (DRP)

The DRP is introduced as a vital enhancement to the RAP. This phase specifically addresses the challenge of uneven workload distribution among UAVs, where certain UAVs are burdened with disproportionately long tour times. Such imbalances in the initial region allocation can lead to inefficiencies in the overall search operation. To counter this, the DRP implements an iterative process that not only adjusts the area allocated to each UAV for a more uniform distribution but also refines the paths within these areas. This dual strategy, encompassing both the dynamic reallocation of search areas and the subsequent optimization of paths via ACO, ensures a more effective and balanced utilization of UAV resources.

4.2.1 Dynamic reallocation algorithm

The Dynamic Reallocation algorithm corrects imbalances in the initial UAV coverage sequences. It begins by identifying the UAV with the longest tour time, which we designate as the Source UAV. This UAV is the primary candidate for path modification.

The algorithm operates iteratively by selecting eligible nodes from the Source UAV's path for reassignment. These candidate nodes are limited to those within the current priority region being optimized to maintain the integrity of paths established for higher-priority regions.

Repathing involves transferring a candidate node from the Source UAV to the path of another, termed the For the Destination UAV, the node can be integrated at any point that maintains path continuity. Conversely, the Source UAV's path must also remain continuous after the node is relinquished.

Furthermore, the algorithm incorporates a hyperparameter m, which dictates the number of attempts for dynamic reallocation. The process attempts up to the minimum of the number of candidate nodes for reallocation and m. After these attempts, the algorithm checks for an improvement in the objective function (*i.e.*, a reduction in the maximum tour length). If no improvement is observed, the algorithm concludes that the dynamic reallocation has failed. This scenario then triggers the termination of the algorithm, since continued attempts are unlikely to yield better results.

This detailed approach in the Dynamic Reallocation Algorithm ensures a more equitable distribution of workload among UAVs. By meticulously repathing nodes and considering the impact on the overall objective function, the algorithm plays a pivotal role in enhancing the efficiency of the UAV network. The complete process of the dynamic reallocation algorithm is presented in Algorithm 3-1 as a pseudo-code.

Input: Path set $P_{\mathcal{U}}$
Output: New Path set $\tilde{P}_{\mathcal{U}}$

$T_{\mathcal{U}} \leftarrow$ tour length set of P_u
$u_{source} \leftarrow \arg\max(T_{\mathcal{U}})$ /* *Identify the Source UAV* */
$CandidateNodes \leftarrow \forall\, n \in P_{u_{source}}$ that *can be re-pathed*
$trial \leftarrow \min(\text{len}(CandidateNodes), m)$
repeat
    $trial \leftarrow trial - 1$
    $target_node \leftarrow$ select a random node from $CandidateNodes$
    $u_{dest} \leftarrow$ select a Destination UAV
    $\tilde{P}_{\mathcal{U}} \leftarrow re\text{-}path(target_node, u_{source}, u_{dest})$
    $\tilde{T}_{\mathcal{U}} \leftarrow$ tour length of $\tilde{P}_u$
    if $\max(\tilde{T}_{\mathcal{U}}) < \max(T_{\mathcal{U}})$ **then**
        break
    else
        $CandidateNodes \leftarrow CandidateNodes \setminus \{target_node\}$
    end if
until $trial = 0$

Algorithm 3-1 Dynamic Reallocation Algorithm

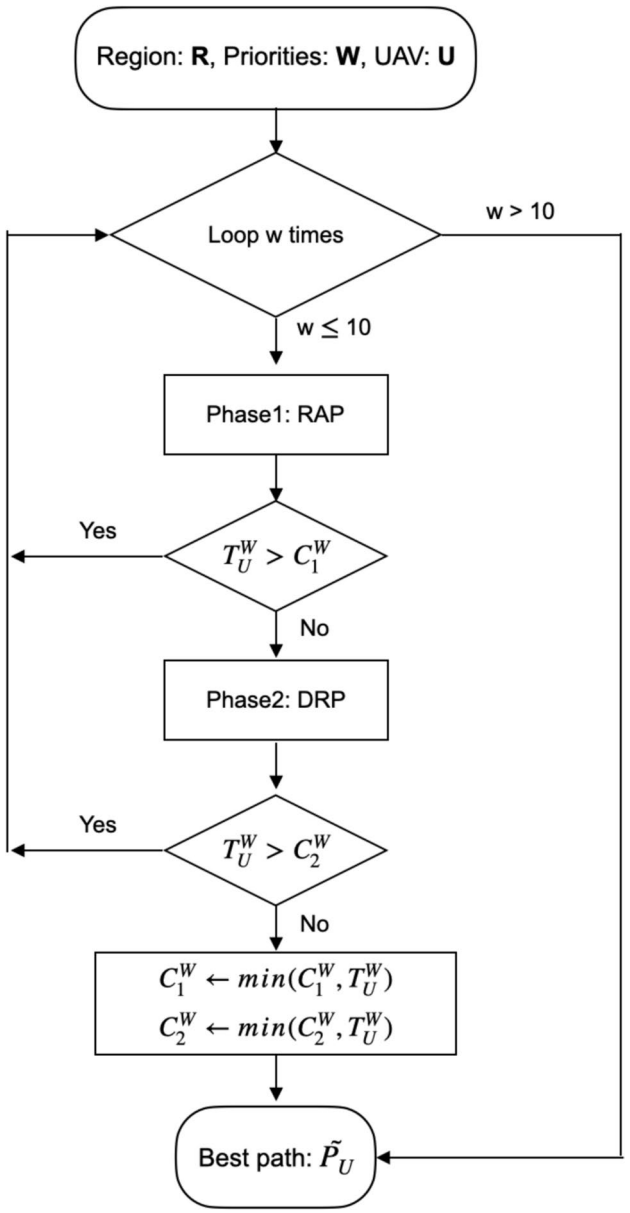


Fig. 2 Flowchart of Dynamic Allocation Algorithm

4.2.2 Ant Colony Optimization (ACO)

Parallel to dynamic allocation, the ACO algorithm is employed for further refinement of each UAV's path within their allocated regions. The constructive search process of ACO is particularly suitable for combinatorial path planning problems with sparse feasible routes, since pheromone updates effectively guide exploration away from dead-ends toward promising paths. Moreover, the pheromone–heuristic mechanism naturally accommodates dynamic regional weights, allowing priority adjustments to be seamlessly integrated into the optimization. This makes ACO well aligned with the requirements of SAR operations where

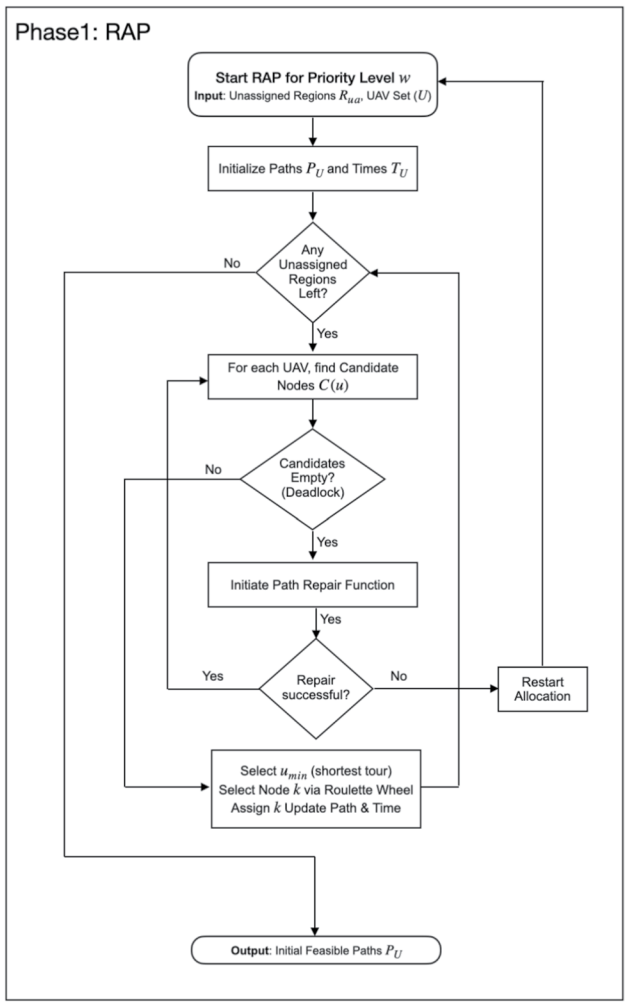


Fig. 3 Flowchart of RAP

time-sensitive priorities are critical. The iterative refinement through ACO also enables potential subsequent dynamic reallocations, facilitating a robust optimization process that adapts to changing mission conditions.

4.2.2.1 Region Selection and Dead-lock Handling The Region Selection in ACO specifically targets a set of weighted nodes, ensuring that previously established paths of different weights remain unaltered. This design maintains the integrity of the optimization while reflecting area priorities. In addition, the population-based nature of ACO supports distributed coordination among heterogeneous UAVs, making it advantageous compared with other metaheuristics that often require centralized synchronization.

In ACO, the region selection process for each UAV is based on a set of probabilistic rules, incorporating both pheromone information, $\tau(i, j)$, and heuristic desirability, $\eta(i, j)$. The pheromone trails represent the learned experience of the ants, while the heuristic information typically

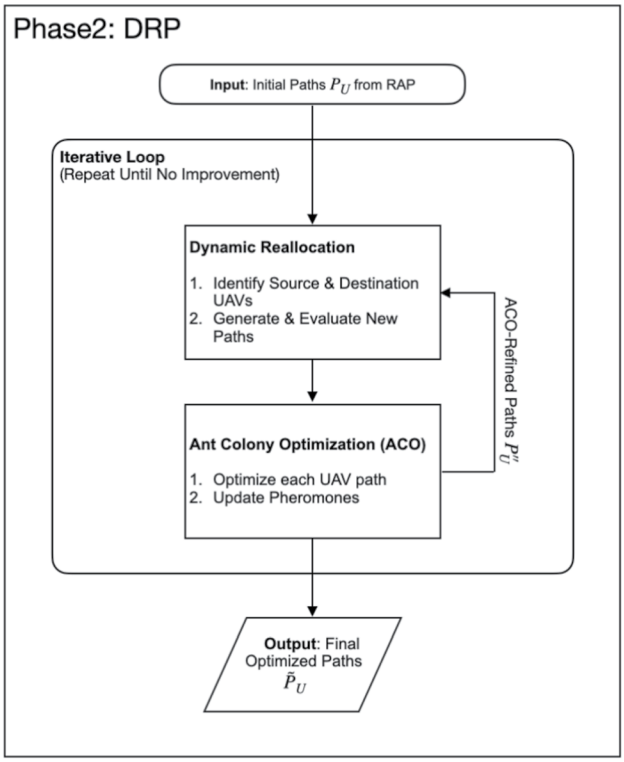


Fig. 4 Flowchart of DRP

inversely relates to the time between regions. The probability of an ant moving from region i to j is influenced by predefined parameters: α ($\alpha > 0$) and β ($\beta > 0$), which control the effect of pheromone and heuristic information,

Table 2 Performance of UAVs

UAV	Velocity (m/s)	Angular velocity (rad/s)	Battery time (min)	Camera footprint (m^2)
1	4	0.5	30	20×20
2	4	0.5	30	20×20
3	4	1	30	20×20

respectively. Additionally, q_0 ($0 \leq q_0 \leq 1$) regulates the balance between exploitation and exploration behaviors (Liu et al. 2016).

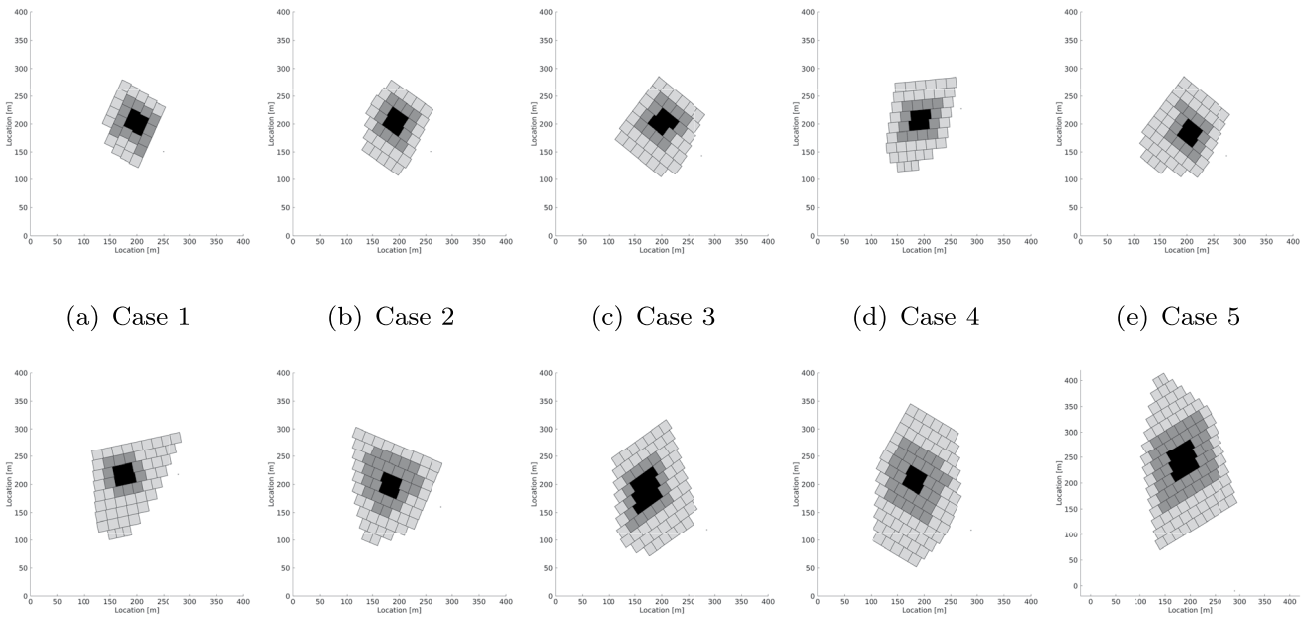
The selection is determined by:

$$R_k = \begin{cases} \arg\max \tau(i,j)^{\alpha}\eta(i,j)^{\beta}, & \text{if } q \leq q_0 \\ \text{Roulette Wheel Selection}, & \text{otherwise} \end{cases} \tag{18}$$

For Roulette Wheel Selection, the probability $p(i,j)$ is computed as:

$$p(i,j) = \begin{cases} \frac{\tau(i,j)^{\alpha}\eta(i,j)^{\beta}}{\sum_{k\in\Pi_{\alpha}}\tau(i,k)^{\alpha}\eta(i,k)^{\beta}}, & \text{if } R_j \in \Pi_{\alpha} \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

where Π_{α} represents the set of regions that an ant can currently move to. In scenarios where ants encounter dead-lock due to the lack of adjacent regions for movement, the repair function as outlined in Algorithm 2-2 is utilized. The number of repair attempts is limited to the hyperparameter p, and if these attempts fail, the process initiates t times, as defined by another hyperparameter. In the event of initialization



(a) Case 1 (b) Case 2 (c) Case 3 (d) Case 4 (e) Case 5

(f) Case 6 (g) Case 7 (h) Case 8 (i) Case 9 (j) Case 10

Fig. 5 Illustration of test cases

Table 3 Impact of the parameter ρ on the objective value and computational time

Case	ρ=0.1		ρ=0.3		ρ=0.5		ρ=0.7		ρ=0.9	
	Obj.value	*Time*	*Obj.value*	*Time*	*Obj.value*	*Time*	*Obj.value*	*Time*	*Obj.value*	*Time*
1	84.67	3.98	97.71	1.2	84.4	1.43	105.47	1.95	111.61	2.05
2	101.3	17.37	116.87	2.01	113.84	5.68	112.27	2.03	119.28	3.47
3	119.95	19.09	121.77	4.65	132.19	4.27	125.82	2.56	137.54	1.61
4	118.13	18.62	125.14	4.48	127.34	10.62	142.93	6.22	125.44	3.57
5	131.97	20.58	151.52	7.05	142.24	51.15	148.77	20.43	144.76	21.44
6	151.92	27.43	149.34	11.79	165.62	47.88	162.99	27.91	153.22	20.49
7	167.8	119.3	183.07	8.79	216.8	57.57	221.24	123.52	187.45	88.89
8	189.69	229.46	215.34	85.18	226.72	170.23	207	64.99	196.41	161.64
9	246.59	391.09	289.41	1682.46	307.99	1697.6	261.32	4105.29	241.9	232.33
10	321.89	444.26	315.96	39.72	362.33	15.32	486.93	0.25	323	270.2

Obj.value = Completion time (sec), *Time* = Computational time (sec)

Table 4 Impact of the parameter p on the objective value and computational time

Case	$p=1$		$p=3$		$p=5$		$p=7$	
	Obj.value	*Time*	*Obj.value*	*Time*	*Obj.value*	*Time*	*Obj.value*	*Time*
1	89.99	12.97	86.73	12.21	84.67	3.98	83.11	22.14
2	105.58	16.32	106.03	10.05	101.30	17.37	107.64	15.12
3	117.76	19.58	120.08	17.91	119.95	19.09	118.44	13.97
4	120.09	18.15	121.76	17.63	118.13	18.62	125.04	25.55
5	137.95	42.73	133.25	36.15	131.97	20.58	139.12	31.78
6	148.96	30.76	158.15	62.08	151.92	27.43	152.95	24.67
7	172.73	54.03	174.14	38.86	167.80	119.30	181.60	76.67
8	192.56	125.97	186.19	180.92	189.69	229.46	196.58	57.64
9	229.61	943.82	231.86	1000.66	246.59	391.09	247.54	1692.73
10	335.34	506.25	300.78	171.32	321.89	444.26	333.21	639.59

Obj.value = Completion time (sec), *Time* = Computational time (sec)

Table 5 Experimental results comparing the proposed method and a commercial solver

Case	Commercial solver		Proposed method		Optimality
	Obj.value	*Time*	*Obj.value*	*Time*	gap (%)
1	84.63	26.01	84.67	3.98	0
2	95.33	85.76	101.3	17.37	6.3
3	109.61	220.43	119.95	19.09	9.4
4	112.92	577.23	118.13	18.62	4.6
5	120.28	559.66	131.97	20.58	9.7
6	135.96	4770.71	151.92	27.43	11.7
7	158.67	2305.05	167.8	119.3	5.8
8	169.05	2597.09	189.69	229.46	12.2
9	-	>15000	246.59	391.09	
10	-	>15000	321.89	444.26	

Obj.value = Completion time (sec), *Time* = Computational time (sec)

Table 6 Experimental results comparing the proposed method and a reinforcement learning approach

Case	Reinforcement learning (Ai et al. 2021)	Proposed method
	Obj.value	*Obj.value*
1	84.64	84.67
2	103.77	101.3
3	113.67	119.95
4	122.02	118.13
5	130.98	131.97
6	145.46	151.92
7	174.81	167.8
8	199.54	189.69
9	277.11	246.59
10	313.94	321.89

Obj.value = Completion time (sec)

fail t times without successfully establishing a path, the ant reverts to its initial path.

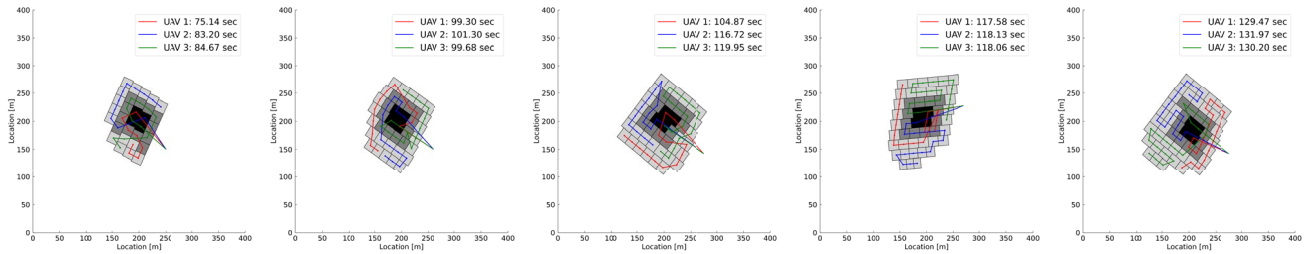
4.2.2.2 Pheromone update The initial pheromone matrix and heuristic information are set for all regions:

$$\tau(i,j) = \tau_0 = \frac{1}{m \cdot L}, \quad \eta(i,j) = \frac{1}{T_{i,j}}, \quad \forall i,j \in R \tag{20}$$

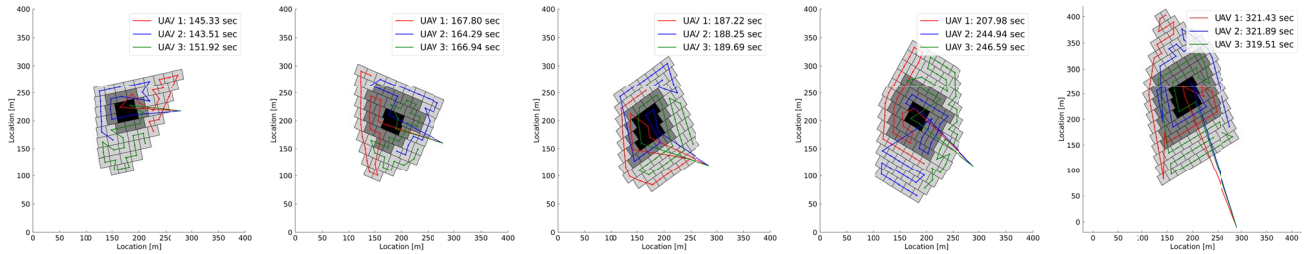
where L is the upper bound (i.e., initial travel time of the UAV), and m represents the number of nodes each ant must visit. The heuristic information $\eta(i,j)$ is defined as the inverse of the time between regions, emphasizing closer regions as more desirable.

Local pheromone updates occur after each ant's move:

$$\tau(i,j) = (1-\rho)\tau(i,j) + \rho\tau_0 \tag{21}$$



(k) Case 1 (l) Case 2 (m) Case 3 (n) Case 4 (o) Case 5



(p) Case 6 (q) Case 7 (r) Case 8 (s) Case 9 (t) Case 10

Fig. 6 Illustration of results for test cases

Table 7 Comparison of Objective Values for RAP vs. RAP + DRP

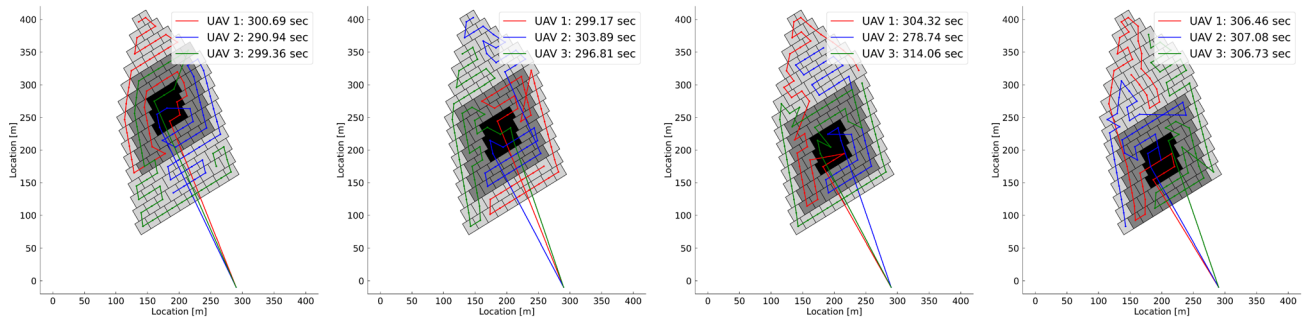
Case	RAP	RAP + DRP	Improvement (%)
1	105.37	84.67	19.65%
2	125.15	101.30	19.06%
3	137.75	119.95	12.92%
4	164.47	118.13	28.18%
5	162.84	131.97	18.96%
6	175.69	151.92	13.53%
7	244.52	167.80	31.38%
8	312.31	189.69	39.26%
9	279.61	246.59	11.81%
10	400.55	321.89	19.64%
Average	**210.83**	**163.39**	**22.44%**

where ρ $(0 < \rho < 1)$ is a predefined decay parameter. This local update reduces the pheromone trail, preventing overemphasis on previously visited paths and encouraging exploration.

Global updates are applied after all ants complete their tours, adjusting the pheromone trails based on the quality of the paths found:

$$\tau(i,j) = (1-\epsilon)\tau(i,j) + \epsilon\Delta\tau(i,j) \tag{22}$$

where ϵ $(0 < \epsilon < 1)$ is another decay parameter. The adjustment $\Delta\tau(i,j)$ is defined as:



(a) Case 10-1 (b) Case 10-2 (c) Case 10-3 (d) Case 10-4

Fig. 7 Illustration of results for Case 10 with changed locations of regional priorities

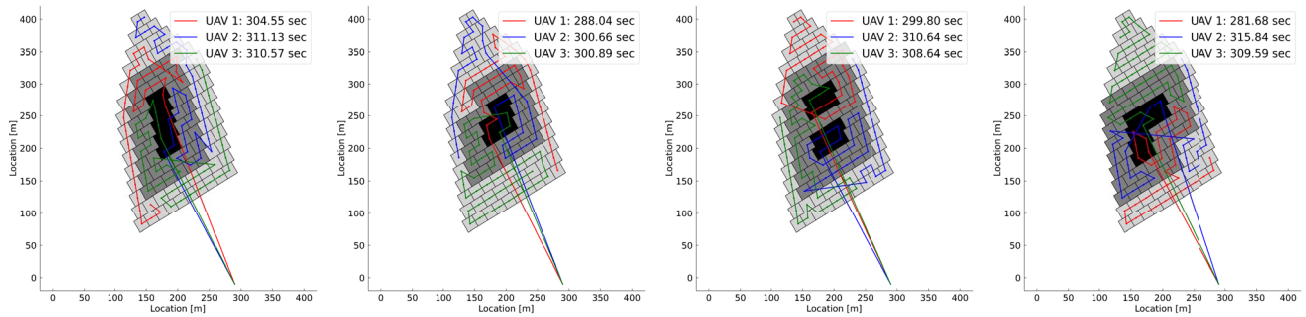
$$\Delta\tau(i,j) = \begin{cases} \frac{1}{\tilde{L}}, & \text{if } (R_i, R_j) \in \tilde{P} \\ 0, & \text{otherwise} \end{cases} \quad (23)$$

Here, $\tilde{L}$ is the time of $\tilde{P}$, the globally best path found so far. Global updates enhance pheromone levels on the arcs of $\tilde{P}$, reinforcing efficient paths, while reducing them on others. This global update rule accelerates convergence by effectively distributing knowledge from successful routes. The algorithm continues until the predetermined number of evolutionary generations is reached. The complete process of the ACO is presented in Algorithm 3-2 as a pseudo-code.

Input: Path P of UAV u, ACO parameters
Output: Optimized Path $\tilde{P}$ of UAV u

$R \leftarrow$ Set of regions in Path P_w
$L \leftarrow$ Tour length of Path P
$\forall i, j \in R$, $\tau(i,j) \leftarrow \frac{1}{m \cdot L}$, $\eta(i,j) \leftarrow \frac{1}{T_{i,j}}$
for $g = 1$ to G **do**
  for $a = 1$ to A **do**
    $R_i \leftarrow$ Place the ant a on $P_{w+1}[-1]$
    $\Pi_a \leftarrow R$
    repeat
      if initialization $= t$ **then**
        $\tilde{P}_a \leftarrow P$
        break
      end if
      $R_i \leftarrow$ Current region in which a is placed
      if Adj$(R_i) = \emptyset$ **then**
        Try *repair* for maximum p times
        If fail, initialize the path of the ant a
      end if
      $q \leftarrow$ random number in $[0, 1]$
      $R_k \leftarrow$ the next region selected via Eq. (18)
      Move the ant a from R_i to R_k
      Remove R_k from Π_a, Update $\tilde{P}_a$
      local pheromone via Eq. (21)
    until $\Pi_a = \emptyset$
  end for
  $\tilde{P} \leftarrow \max_a(\tilde{P}_a)$
  global pheromone update via Eq. (22)
end for

Algorithm 3-2 ACO Algorithm



(a) Case 10-5 (b) Case 10-6 (c) Case 10-7 (d) Case 10-8

Fig. 8 Illustration of results for Case 10 with changed shapes of regional priorities

4.2.3 Iterative optimization process

The dynamic allocation phase incorporates a strategic iterative process that alternates between Dynamic Reallocation and ACO. Initially, the process focuses on Dynamic Reallocation, attempting to redistribute UAV paths for more balanced coverage. This reallocation is pursued until it is no longer feasible to make further improvements.

Upon reaching the limit of dynamic reallocation's effectiveness, the process shifts to the ACO algorithm. ACO refines the UAV paths within their assigned regions, optimizing the coverage sequence based on heuristic and pheromone information. Following the application of ACO, the process evaluates the potential for additional dynamic reallocations.

This iterative cycle between dynamic reallocation and ACO continues, progressively enhancing path efficiency. The stopping criterion for this iterative process is determined by the inability of dynamic reallocation to yield any improvements during a cycle. This indicates that no further adjustments can be made to the paths through reallocation. When this scenario occurs, the algorithm concludes, and the paths optimized by the last application of ACO are considered final. The complete process of the DRP is presented in Algorithm 3 as a pseudo-code. We provide the flow chart of the proposed method in Fig. 2 and the deatile version of RAP and DRP module flowchart in Fig. 3 and Fig. 4 respectively.

Input: Path set $P_{\mathcal{U}}$
Output: New Path set $\tilde{P}_{\mathcal{U}}$
$\tilde{P}_u \leftarrow P_{\mathcal{U}}$
repeat
  $dynamic_iter \leftarrow 0$
  repeat
    $\tilde{P}_u \leftarrow Reallocation(\tilde{P}_u)$
    $dynamic_iter \leftarrow dynamic_iter + 1$
  until failure
  if $dynamic_iter = 0$ **then**
    break
  else
    for u in $\mathcal{U}$ **do**
      $\tilde{P}_u \leftarrow ACO(\tilde{P}_u)$
    end for
  end if
until

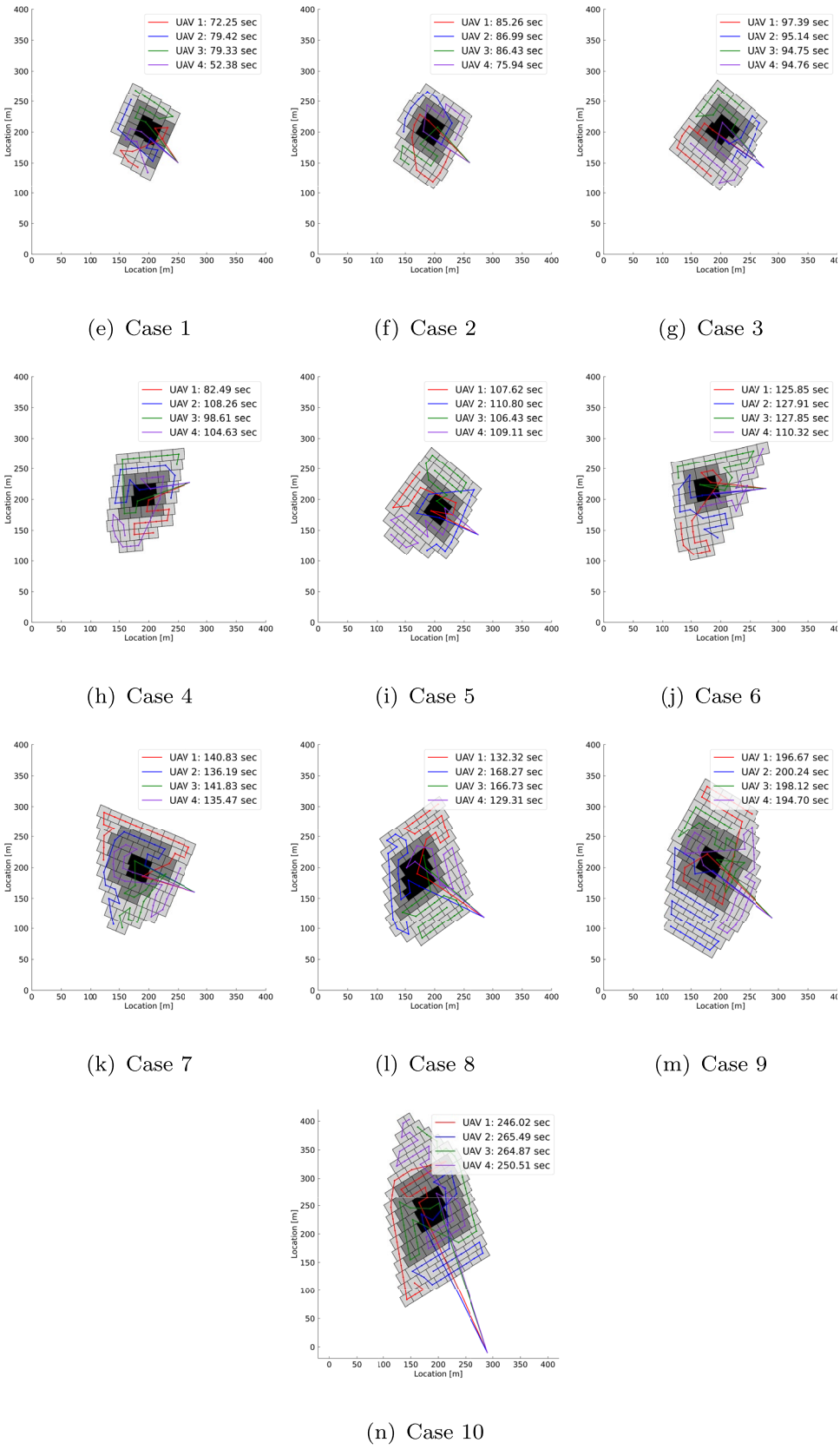
Algorithm 3 Dynamic Refinement Phase (DRP)

Table 8 Impact of the number of UAVs on the objective value and computational time

Case	2 UAVs		3 UAVs		4 UAVs		5 UAVs	
	Obj.value	*Time*	*Obj.value*	*Time*	*Obj.value*	*Time*	*Obj.value*	*Time*
1	117.17	10.45	84.67	3.98	79.42	26.33	63.53	26.18
2	129.78	13.55	101.30	17.37	86.99	31.42	74.33	32.49
3	171.92	17.96	119.95	19.09	97.39	32.18	86.87	32.61
4	166.18	22.89	118.13	18.62	108.26	32.59	86.15	30.12
5	189.00	35.08	131.97	20.58	110.80	33.65	103.45	32.91
6	219.67	104.07	151.92	27.43	127.91	34.20	110.63	36.08
7	247.07	23.10	167.80	119.30	141.83	38.60	118.26	35.14
8	295.11	99.64	189.69	229.46	168.27	30.56	133.09	14.53
9	371.42	566.60	246.59	391.09	200.24	507.61	190.76	38.86
10	424.40	1796.95	321.89	444.26	265.49	645.37	234.24	49.22

Obj.value = Completion time (sec), *Time* = Computational time (sec)

Fig. 9 Illustration of results with 4 UAVs



(e) Case 1 (f) Case 2 (g) Case 3

(h) Case 4 (i) Case 5 (j) Case 6

(k) Case 7 (l) Case 8 (m) Case 9

(n) Case 10

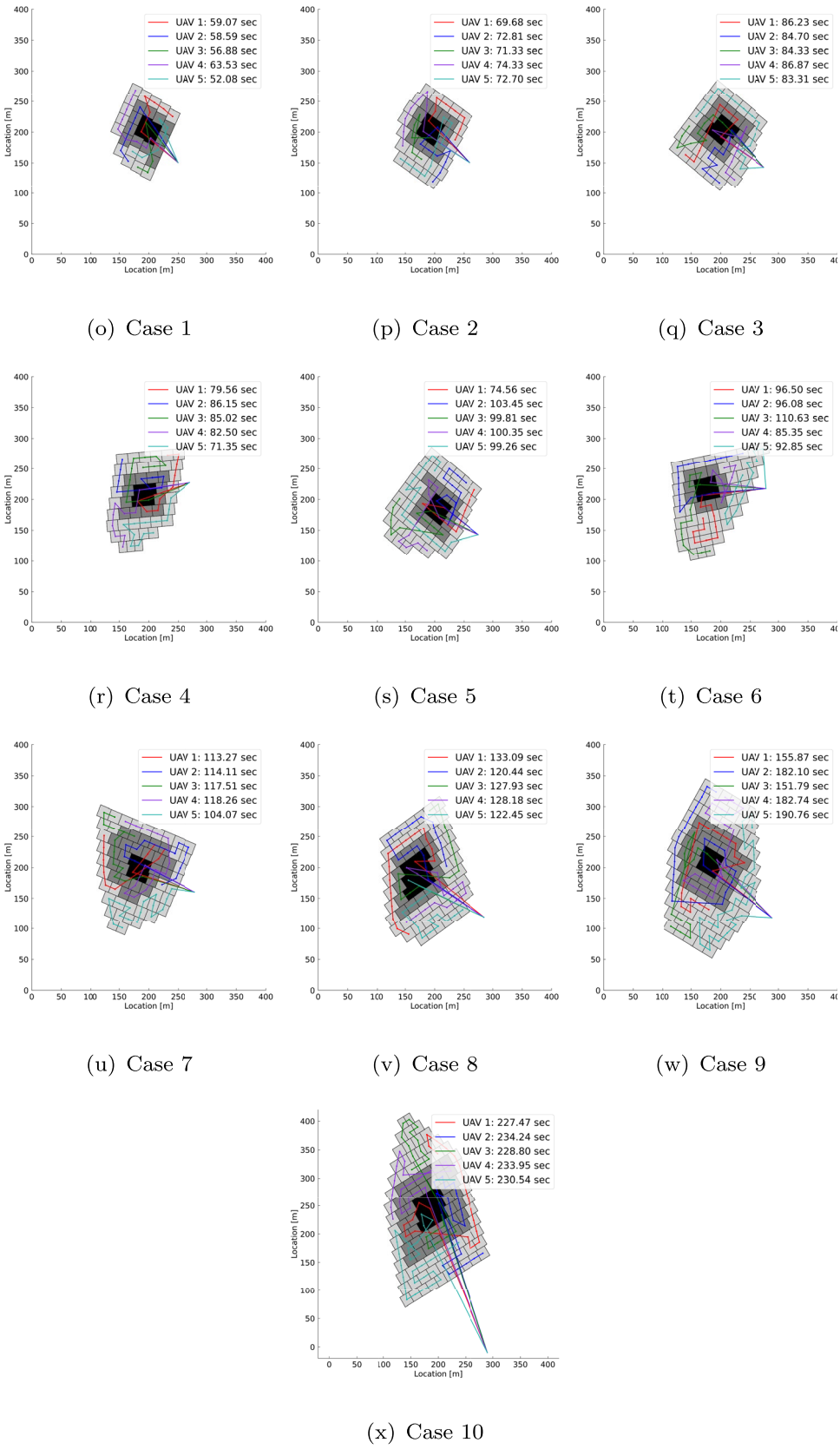


Fig. 10 Illustration of results with 5 UAVs

(o) Case 1 (p) Case 2 (q) Case 3

(r) Case 4 (s) Case 5 (t) Case 6

(u) Case 7 (v) Case 8 (w) Case 9

(x) Case 10

5 Numerical experiments

In this section, numerical experiments are conducted to demonstrate the effectiveness of our proposed Dynamic Allocation Algorithm for maritime SAR operations. We present the results of the three experiments as follows. First, the proposed method involves various hyperparameters. In particular, since ρ and p directly affects the performance of the proposed method, we focus on the change in ρ and p. Second, we identify the gap with the optimal solution to verify the performance of the proposed algorithm. We employed the mathematical model presented in (Cho et al. 2021) and utilized the commercial solver Gurobi 10.0.2 to derive the optimal solution. Additionally, we compare the performance of our method with the reinforcement learning (RL) approach proposed by (Ai et al. 2021), which represents a recent machine learning-based solution to the coverage problem. Third, we demonstrate the robustness and practicality of the proposed algorithm through experiments that involve the transformation of the location and shape of regional priorities. The program was implemented on a MacBook Pro with an Apple M1 Pro chip, 8 cores, and 16 GB of memory.

5.1 Experimental setup

Numerical experiments are conducted using 10 test cases derived from actual sea coordinates off Gijang-eup, Gijang-gun, Busan, Republic of Korea. These coordinates are taken from publicly available maritime charts (not SAR operation logs) and discretized into grid structures for the experiments. For each case, the grid structure was generated using the area minimization algorithm detailed in Section 3, resulting in irregular search areas with node counts ranging from 31 to 112. The 10 test instances are classified by size into three categories: **Small** (Case 1, with 31 nodes), **Medium** (Cases 2–6, with 39–57 nodes), and **Large** (Cases 7–10, with 67–112 nodes). The regional priorities were assigned using the concentric layer model formally defined in Equation (5). To ensure algorithmic robustness, the priority band thresholds (k_H and k_M) were varied for each test case. The final configurations of all 10 test scenarios are illustrated in Fig. 5. As detailed in Table 2, three types of UAVs are utilized during the search operation.

Following extensive preliminary experiments involving various parameter combinations, we set the values of 5 and 10 for m and t respectively. For ACO parameters, the number of generations (G) and ants (A) were set to 10 and 30, respectively. The other control parameters—q_0, α, β, and ϵ—were set to moderate values of 0.7, 1, 2, and 0.5, respectively, which ensure a balance between search exploration and exploitation.

5.2 Sensitivity analysis

In this section, objective value and computational time according to the change in the parameter ρ and p of the proposed methodology is presented.

ρ affects the update of pheromones in ACO, and plays a crucial role in the movement of UAVs at each node. We conducted experiments by changing the ρ from 0 to 1, and the results for 0.1, 0.3, 0.5, 0.7 and 0.9 with significant changes are presented in Table 3. When ρ is set to a two-end value, pheromones converge effectively to derive a high-quality solution and reduce computational time. In particular, we observe that in Case 9, the proposed algorithm is underfitted at ρ of 0.3, 0.5, and 0.7, resulting in a significant increase in computational time for solution derivation. Considering the experimental results from this perspective, it can be concluded that our test cases yield a good solution overall when ρ is set to 0.1.

Additionally, we conducted sensitivity analysis on the parameter p, which defines the maximum number of path repair attempts before initiating a restart in the region allocation process. Experiments were performed with p ranging from 1 to 7, and results for 1, 3, 5, and 7 are summarized in Table 4. The analysis indicates that while lower values of p generally result in reduced computational time but may compromise solution quality, whereas higher values can improve solution quality at the expense of increased computation. Setting $p = 5$ provides an effective balance between solution quality and computational efficiency across most test cases. For instance, in Case 1, $p = 5$ yields the best objective value of 84.67 with reasonable computational time. Similarly, for larger instances such as Cases 9 and 10, it provides competitive results within acceptable time limits. Accordingly, $p = 5$ is adopted for the experiments, as it consistently delivers good-quality solutions within acceptable computational times.

5.3 Performance comparison with existing methods

In this section, we present experimental results to validate the performance of the proposed methodology. We compare the objective value of the proposed methodology with the optimal solution obtained from the MILP model proposed by (Cho et al. 2021). Due to the complexity of the model, we set the maximum computation time to 15,000 seconds. We also compare the performance of our method with the RL approach presented in (Ai et al. 2021). Since the original method was designed for a single agent, it was modified to accommodate multiple agents in order to ensure a fair comparison. Furthermore, as the grid maps generated in this study are not based on regular square grids, the Q-value

indexing scheme varied across cases, necessitating separate training for each test instance.

Table 5 shows that the proposed method performs with an average optimality gap of 7.5% compared to the commercial solver. Furthermore, significant improvements are observed with an average computational time of 91.8%. As shown in the results of Cases 9 and 10, the MILP model is unable to derive exact solutions due to the complexity of the problem. In contrast, our proposed method provides results within reasonable computational times. Figure 6 displays the path generated by the proposed method in each case.

Table 6 shows a comparative analysis of the proposed method and the RL approach introduced by Ai et al. (2021). The proposed method achieves superior results in 6 out of 10 cases, with particularly notable improvements in Cases 9 and 10, where completion times are reduced by 11.0% and 2.5%, respectively. As the number of nodes in the graph increases, the state space of the RL approach grows exponentially, and the interactions among UAVs become more complex. These factors likely degrade the learning performance in large-scale scenarios. In contrast, the proposed method consistently delivers competitive and often improved results without relying on extensive training, demonstrating its robustness and scalability in complex environments.

5.4 Impact of the dynamic refinement phase

To quantify the contribution of the algorithm's core components, we conducted an ablation study comparing the performance of the Region Allocation Phase (RAP) alone against the full, two-phase algorithm (RAP + DRP). The data for this analysis is sourced from the same experimental runs used for the performance comparison in Section 5.3. The RAP, being a greedy construction heuristic, is designed to quickly generate a feasible initial solution. The "RAP alone" results represent the objective value of this initial solution, captured just before the DRP begins its optimization process. The RAP + DRP results are the final objective values after the refinement is complete.

The results of this comparison are presented in Table 7. The data clearly demonstrates that the DRP provides a significant and consistent improvement across all test cases, with an average reduction in completion time of 22.44%. This substantial improvement highlights the critical role of the DRP. The greedy nature of the RAP, while fast, is prone to creating imbalanced workloads where one UAV may be assigned a disproportionately long tour, thereby increasing the overall makespan. The DRP is specifically designed to correct this. Its dynamic reallocation mechanism effectively balances the load among UAVs by reassigning nodes from the most burdened UAV to others. Subsequently, the ACO-based path refinement optimizes the travel sequence within these newly allocated regions. This two-stage refinement process is directly responsible for transforming a viable but suboptimal initial solution into the high-quality, balanced final solution. Therefore, this analysis confirms that the DRP is an indispensable component of the proposed methodology, justifying the algorithm's two-phase, construction-plus-improvement architecture.

5.5 Search path according to changes in priorities

In this section, we present experimental results of algorithms applied to different types of regional priorities. For Case 10, experiments were conducted to move the location of regional priorities, as shown in Fig. 7, and to change the shape of regional priorities, as shown in Fig. 8. The results show that the proposed methodology yields the solution to irregular forms of seas generated by currents, waves, and wind.

5.6 Performance according to the number of UAVs

In this section, we present experimental results of the algorithm across varying numbers of UAVs. The experiments were conducted using the same test cases as in previous sections, with UAV configurations ranging from 2 to 5 UAVs. For each configuration, one uav were configured same with UAV 3 and other(s) were identical with UAVs 1 and 2, detailed in Table 2.

Table 8 shows the objective values and computational times for different numbers of UAVs across all test cases. The results demonstrate that as the number of UAVs increases, the objective value significantly decreases, indicating more efficient area coverage. Furthermore, despite the increased complexity of coordinating more UAVs, the solution time tend to decrease as the number of UAVs increases, particularly for larger problem instances such as Cases 9 and 10. For Case 9, the computational time reduced dramatically from 566.60 seconds with 2 UAVs to just 38.86 seconds with 5 UAVs, representing a 93.1% reduction. Similarly, for Case 10, the time decreased from 1796.95 seconds to 49.22 seconds, a remarkable 97.3% reduction. This trend suggests the scalability of the algorithm for large-scale problems where deploying multiple UAVs is beneficial. The phenomenon can be attributed to the efficient task allocation mechanism in the DRP, which effectively distributes the search paths among available UAVs, thereby reducing the computational burden of exploring lengthy individual paths. The results using 4 and 5 UAVs are shown in Fig. 9 and Fig. 10.

6 Conclusion

The escalating number of maritime transport vessels has led to an increased frequency of ship distress accidents, emphasizing the critical need for efficient maritime SAR operations. This paper proposes a novel approach employing a CPP framework and an Dynamic Allocation Algorithm for multiple UAVs in SAR missions, aiming to minimize search time and prioritize areas with a higher likelihood of survivor presence. The methodology consists of two phases: the RAP and the DRP. In the RAP, it generates a solution by employing a greedy approach, selecting the neighbor node with the minimum travel time from the current node considering regional priorities. In the DRP, encompassing both the dynamic reallocation of search areas and the subsequent optimization of paths via ACO ensures a more effective and balanced utilization of UAV resources.

The verification process involves comprehensive numerical experiments using actual sea coordinates from Busan, Republic of Korea, providing a thorough assessment of the proposed algorithm's effectiveness. Sensitivity analysis on hyperparameters, particularly ρ and p, reveals optimal performance when $\rho = 0.1$ and $p = 5$. In comparative experiments, the proposed method demonstrates clear superiority against both a MILP model and a RL-based approach. Compared to the MILP model, it achieves an average optimality gap of 7.5% and reduces computational time by 91.8%. Against the RL method, the proposed approach outperforms in most cases, particularly in large-scale instances where the RL model suffers from the exponential growth of the state space and complex inter-agent dynamics. The proposed method demonstrates computational robustness and scalability, consistently generating high-quality solutions within practical computation times—even in cases where the MILP model faces challenges due to problem complexity.

Despite these promising results, several limitations should be acknowledged. The framework currently assumes grid-based discretization of the search area, which may not fully capture irregular maritime environments. In addition, the present study does not explicitly consider real-time deployment issues such as communication latency or UAV synchronization delays, which could affect practical performance in field operations. Addressing these limitations constitutes an important direction for future research and will help bridge the gap between simulation and real-world SAR deployment.

Future research could focus on extending to the maritime SAR system that integrates with other robotic technologies such as unmanned surface vehicles. This collaborative approach could contribute to the overall improvement of the proposed methodology, ensuring a comprehensive and efficient response to distress situations in the maritime domain. Additionally, exploring Multi-Agent Reinforcement Learning techniques could offer new insights into improving coordination and decision-making among multiple agents in dynamic and complex maritime environments. Another interesting direction is to extend the current deterministic setting toward stochastic scenarios (e.g., uncertain survivor locations or environmental conditions). Beyond SAR, the framework also has potential applications in other multi-agent domains such as environmental monitoring, precision agriculture, and warehouse logistics. These extensions broaden the applicability of the proposed method and address the evolving challenges of both maritime SAR and other real-world domains.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Human participants and/or animals This research does not involve human participants or animals.

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